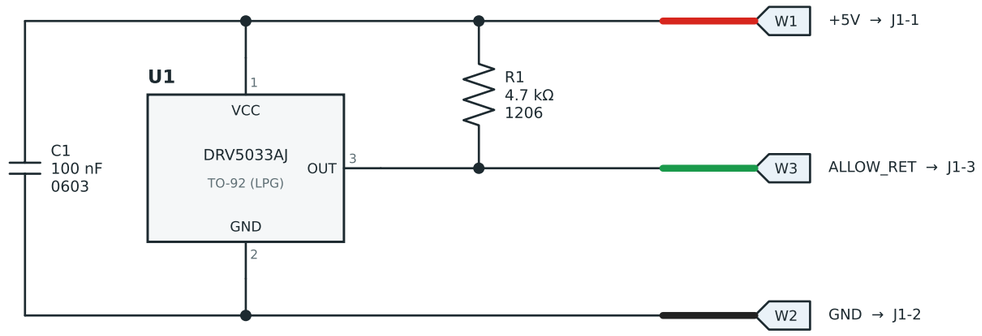
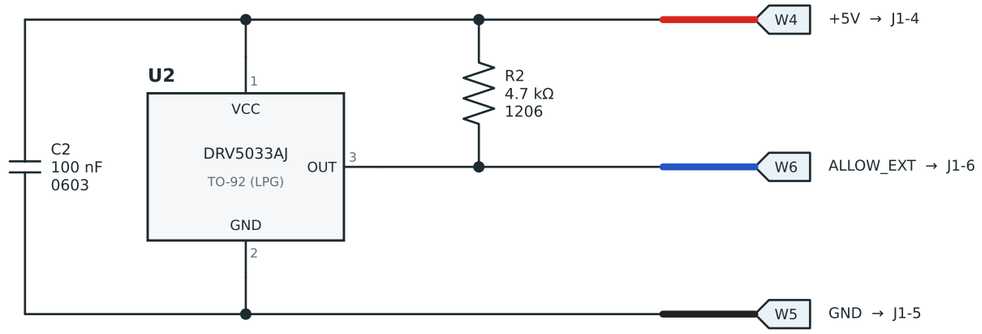
INSIDE THE ACTUATOR

Both Hall end-stop sensors and the N20 motor. Eight wires leave through the rear-cap port and land on J1 of the controller (sheet 2); J1 pin number = wire number.

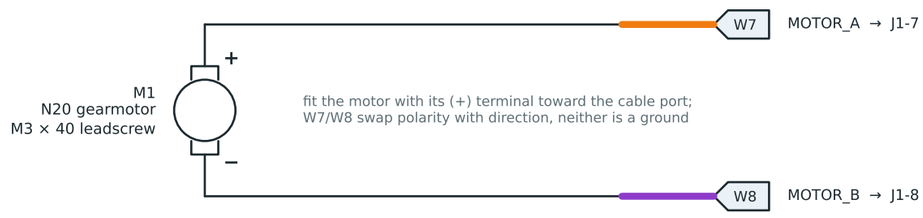
U1 · RETRACT-END HALL SENSOR (MOTOR END OF THE BARREL)



U2 · EXTEND-END HALL SENSOR (HEAD END OF THE BARREL)



M1 · MOTOR (THE UNCHANGED CORE PART)

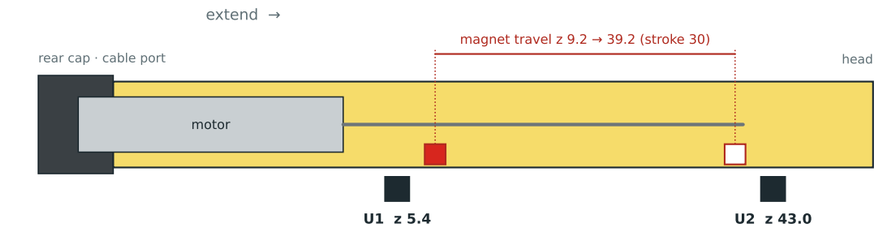


HARNESS · 8-WIRE TAIL IN Ø4.0 / 3.2 MM JACKET

Wire	Colour	Actuator end	J1 pin	Controller net	Ins. Ø	Inside, mm
W1	red	U1 pin 1 VCC	J1-1	+5V	0.4 mm	76
W2	black	U1 pin 2 GND	J1-2	GND	0.4 mm	76
W3	green	U1 pin 3 OUT	J1-3	ALLOW_RET	0.4 mm	76
W4	red	U2 pin 1 VCC	J1-4	+5V	0.4 mm	113
W5	black	U2 pin 2 GND	J1-5	GND	0.4 mm	113
W6	blue	U2 pin 3 OUT	J1-6	ALLOW_EXT	0.4 mm	113
W7	orange	M1 (+)	J1-7	MOTOR_A	0.7 mm	54
W8	purple	M1 (−)	J1-8	MOTOR_B	0.7 mm	62

"Inside" = modelled route from the solder joint to 8 mm past the jacket entry; add the run to the controller and calibration slack. Modelled OD: sensor 0.4 mm (AWG 32 thin-wall PTFE), motor 0.7 mm (AWG 28 PTFE); measure before buying the jacket.

WHERE THE SENSORS SIT · BARREL Z IN MM, 0 = GEARBOX FRONT FACE



Each Hall sits 3.8 mm outboard of the magnet end position because the output switches about 3.9 mm before the magnet centre arrives and releases 4.2 mm after it (typ., magnet model). The sensors slide ±3 mm in the channel for calibration.

HALL SENSOR AND MAGNET

U1, U2	TI DRV5033AJQLPG, omnipolar, open-drain output, 2.5–38 V supply (used at 5 V).
Thresholds	operate BOP 3–12 mT (typ 6.9), release BRP 1–5 mT (typ 3.5). Output LOW = magnet present.
Magnet	N35, Ø4 × 2 mm, magnetised through the 2 mm, axis radial toward the channel, glued in the carrier.
Model	about 98 mT at the sensor; reverse lobe -2.5 mT stays under BOP min, so no early trigger.
Do not	use N42 or stronger: its reverse lobe can reach BOP min and switch the sensor twice.

NOTES

1. Open-drain outputs: R1/R2 pull OUT up to +5 V at the sensor. The 47 kΩ pull-downs on the controller (R3/R4, sheet 2) make a broken signal or supply wire read LOW, which blocks that direction.
2. R1/R2 (1206) and C1/C2 (0603) are soldered straight onto the Hall leads. Sleeve every joint with heat-shrink before the sensor goes into the channel.
3. No splices inside the cylinder: each Hall has its own +5 V and GND wire, joined only at J1 on the controller.
4. Pin numbers are those of the LPG package in the TI datasheet (1 VCC, 2 GND, 3 OUT). Check them on the part, not on the 3D picture.

Inside the actuator

Excavator arm actuator V5 · N20 gearmotor with integrated M3 leadscrew

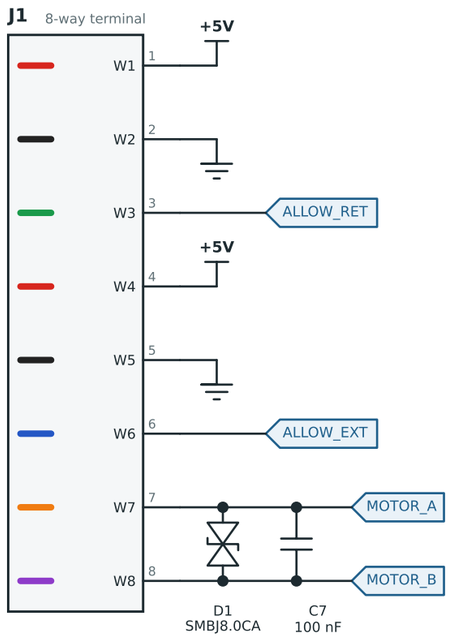
SHEET	REVISION	DATE	FORMAT
1 of 2	A	2026-10-27	A3 landscape

Generated by source/build_schematic.py. Connectivity re-derived from the drawing and checked against the netlist: 8 nets, 16 pins, 8 junctions, no differences.

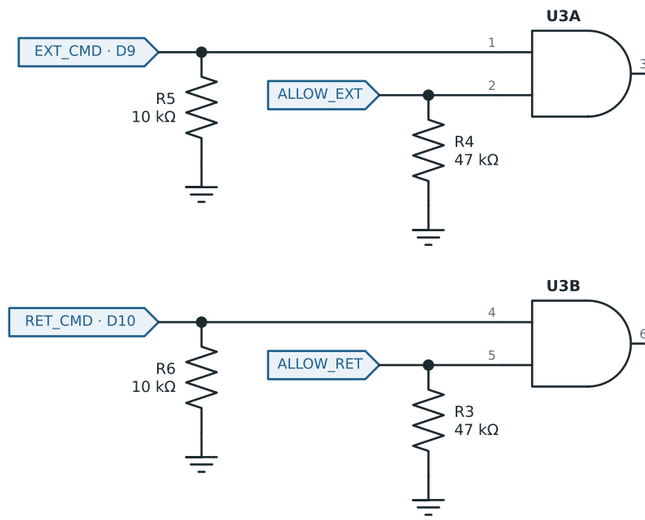
CONTROLLER BOARD

6 V motor supply, 5 V logic. U3 blocks a direction in hardware whenever its Hall output is LOW; the firmware adds current limit, timeout and fault latching.

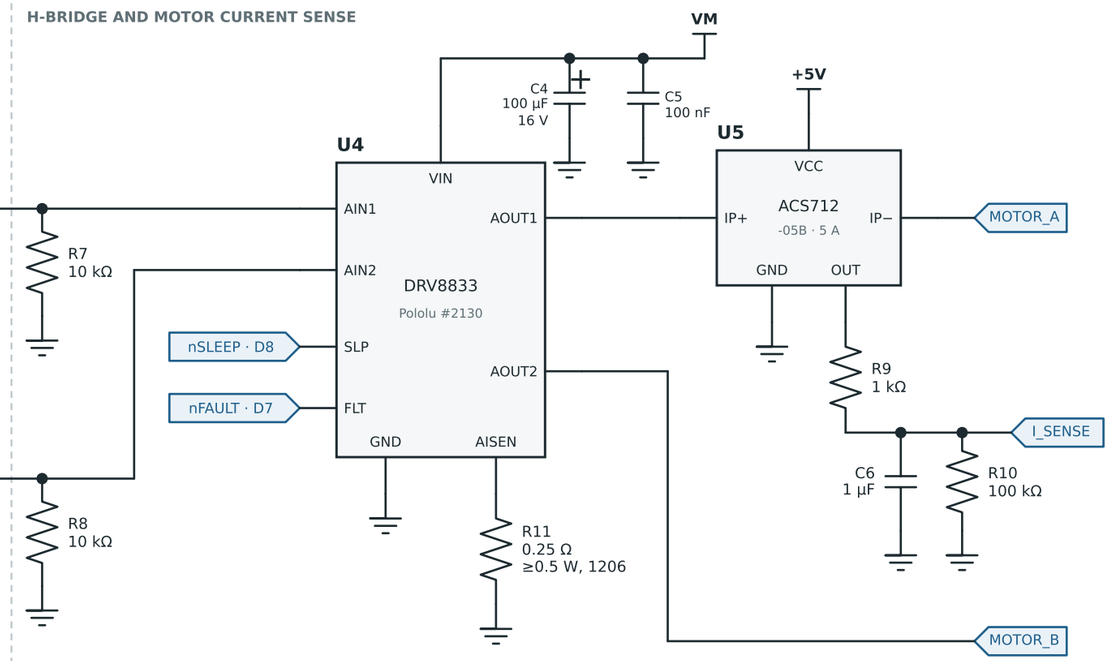
HARNESS FROM THE ACTUATOR



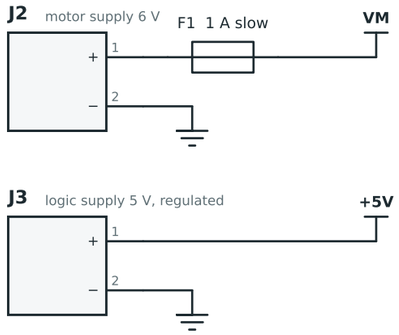
DIRECTION GATES · U3 SN74HC08



H-BRIDGE AND MOTOR CURRENT SENSE

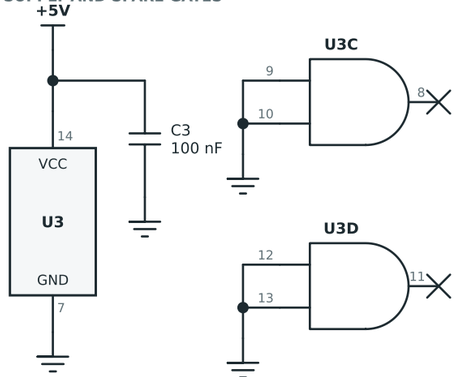


POWER IN



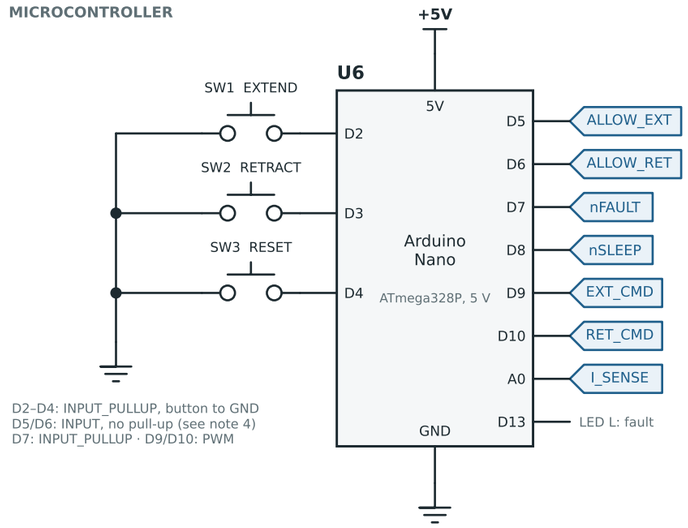
VM: 6 V for the example motor; the DRV8833 works from 2.7 to 10.8 V, so never 12 V. +5V also feeds the Nano 5V pin, U3, U5 and both Hall sensors. All grounds are common.

U3 SUPPLY AND SPARE GATES



Unused inputs tied to GND, outputs 8 and 11 open. C3 sits next to pin 14.

MICROCONTROLLER



D2–D4: INPUT_PULLUP, button to GND
D5/D6: INPUT, no pull-up (see note 4)
D7: INPUT_PULLUP · D9/D10: PWM

DIRECTION LOGIC

Situation	ALLOW_RET	ALLOW_EXT	Extend	Retract
between the ends	H	H	yes	yes
at the retract end (U1)	L	H	yes	blocked
at the extend end (U2)	H	L	blocked	yes
W3 or W1 open	L	H	yes	blocked
W6 or W4 open	H	L	blocked	yes
both LOW (sensor fault)	L	L	blocked	blocked

AIN1 = EXT_CMD · ALLOW_EXT
AIN2 = RET_CMD · ALLOW_RET

Not detected: an open GND wire (W2/W5), a signal shorted to +5 V or a lost magnet. The current limit and the 30 s run timeout are the backup.

NOTES

1. Never feed VM into the Nano 5V pin. If the Nano is on USB, J3 may stay connected (the Nano's USB diode blocks back-feed).
2. Current limit: cut the AISEN-to-GND break point on the back of the Pololu #2130 and solder R11 on its 1206 pads → chopping at about 0.2 V / 0.25 Ω = 0.8 A (0.64–0.96 A). Channel B is unused. SLP is pulled high on the carrier; FLT is open-drain (D7 uses INPUT_PULLUP).
3. U5 must be the ACS712-05B (185 mV/A). The firmware zeroes it at power-up and trips at 550 mA for 150 ms. These are start values: measure the motor.
4. R5–R8 hold the gate inputs and AIN1/AIN2 LOW while the Nano resets, so the motor cannot start by itself. Keep D5/D6 plain INPUT: an internal pull-up would make a broken W3/W6 read HIGH.
5. First power-up at low voltage, rod disconnected: EXT_CMD must extend. If it retracts, swap W7 and W8 at J1.

Controller board

Excavator arm actuator V5 · N20 gearmotor with integrated M3 leadscrew

SHEET	REVISION	DATE	FORMAT
2 of 2	A	2026-10-27	A3 landscape

Generated by source/build_schematic.py. Connectivity re-derived from the drawing and checked against the netlist: 21 nets, 91 pins, 19 junctions, no differences.